1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model that predicts whether an employee is likely to leave the organization (**Attrition**).

Business Context

Employee turnover leads to substantial direct and indirect costs, including recruitment overhead, onboarding expenditure, lost productivity, and diminished team morale. A predictive HR analytics system can:

- **Reduce turnover risk** by identifying employees showing high flight-risk indicators early.
- **Improve workforce retention** through targeted compensation, workload management, and career development initiatives.
- **Optimize HR operations** and minimize financial and institutional knowledge losses.

1.2 Objective

- **Predict employee attrition** (Yes / No).
- **Identify key drivers** contributing to employee turnover (e.g., workload, compensation, job satisfaction, tenure).
- **Assist HR leadership & managers** in proactive decision-making and tailored retention planning.

1.3 Target Variable

- **Attrition**
  - **1 (Yes)** Employee left the company (**237 records, 16.12%**)
  - **0 (No)** Employee retained (**1,233 records, 83.88%**)

1.4 Input Variables

The dataset contains **34 input attributes** categorized across core operational domains:

- **Demographics & Personal:** Age, Gender, MaritalStatus, Education, EducationField
- **Job & Work Profile:** Department, JobRole, JobLevel, BusinessTravel, DistanceFromHome, OverTime
- **Compensation & Financial:** MonthlyIncome, DailyRate, HourlyRate, MonthlyRate, PercentSalaryHike, StockOptionLevel
- **Satisfaction & Engagement (1–4 Scale):** EnvironmentSatisfaction, JobSatisfaction, JobInvolvement, RelationshipSatisfaction, WorkLifeBalance
- **Work History & Tenure:** TotalWorkingYears, NumCompaniesWorked, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager, TrainingTimesLastYear, PerformanceRating
- **Redundant Metadata / Identifiers:** EmployeeCount, Over18, StandardHours, EmployeeNumber

1.5 Initial Data Checks

Performed

- df.shape **1,470 rows** and **35 columns**
- df.info() **26 numerical** (int64), **9 categorical** (object), **0 missing values**
- df.describe() Evaluated summary statistics (mean, median, std, min, max) for all numerical features
- df.head() / df.tail() Inspected structure and sample records

Observations

- **Zero missing values** across the entire dataset.
- **Mix of variable types:** Continuous (e.g., MonthlyIncome), discrete (e.g., YearsAtCompany), ordinal (e.g., JobSatisfaction), and nominal (e.g., JobRole).
- **Class Imbalance:** Target variable Attrition is imbalanced (~16.1% positive class).

1.6 Data Cleaning Summary

Unique & Zero-Variance Columns Removed

- **Constant / Zero-Variance Features:** EmployeeCount (always 1), Over18 (always 'Y'), StandardHours (always 80).
- **Identifier Feature:** EmployeeNumber (1,470 unique sequential IDs with no predictive value).

Null Value Handling

Column Type	**Null Count**	**Strategy**
Numerical (int64)	0	None required (*Median Imputation recommended for future production pipelines*)
Categorical (object)	0	None required (*Mode Imputation recommended for future production pipelines*)

Duplicate Handling

- **Checked using:** df.duplicated().sum() **0 duplicate rows** found.
- **Action:** Strategy established to run df.drop_duplicates() in pipeline preprocessing.

Error Value Handling

- Checked for placeholder/invalid strings (e.g., "?", "unknown", negative values).
- Values were clean and standardized across standard categories.

Column Segregation

- **Categorical Columns (9):** Attrition, BusinessTravel, Department, EducationField, Gender, JobRole, MaritalStatus, Over18, OverTime
- **Numerical Columns (26):** Age, DailyRate, DistanceFromHome, Education, EmployeeCount, EmployeeNumber, EnvironmentSatisfaction, HourlyRate, JobInvolvement, JobLevel, JobSatisfaction, MonthlyIncome, MonthlyRate, NumCompaniesWorked, PercentSalaryHike, PerformanceRating, RelationshipSatisfaction, StandardHours, StockOptionLevel, TotalWorkingYears, TrainingTimesLastYear, WorkLifeBalance, YearsAtCompany, YearsInCurrentRole, YearsSinceLastPromotion, YearsWithCurrManager

1.7 Outlier Detection & Treatment

Technique Used: Interquartile Range (IQR) Method

- {Lower Bound} = Q1 - 1.5 *IQR
- {Upper Bound} = Q3 + 1.5 *IQR

Outlier Distribution Found

- **We did not found any outlier.**

1.8 Exploratory Analysis Summary

Univariate Analysis

- **Income Distribution:** Right-skewed MonthlyIncome with an average of **6,503** (median 4,919, max 19,999).
- **Age Profile:** Normally distributed around a mean age of **36.9 years** (range: 18 – 60).
- **Workload & Travel:** 28.3% of the workforce works OverTime; 71.0% travel rarely, 18.8% travel frequently, and 10.2% do not travel.

Bivariate Analysis

- **Attrition vs. OverTime:** Employees working OverTime have a **30.53%** attrition rate compared to **10.44%** for non-overtime staff.
- **Attrition vs. Income & Age:** Departed employees earn significantly lower average monthly income **4,787** vs. **6,833**) and are younger on average (**33.6** vs. **37.6** years).
- **Attrition vs. Job Role:** Highest attrition rates observed in **Sales Representative (39.76%)**, **Laboratory Technician (23.94%)**, and **Human Resources (23.08%)**, compared to **Manager (4.90%)** and **Research Director (2.50%)**.

- **Attrition vs. Marital Status & Travel:** Single employees (**25.53%**) and Frequent Travelers (**24.91%**) display noticeably higher churn rates.

Key Observations

- **Overtime** and **Lower Monthly Compensation** are the strongest risk indicators for employee attrition.
- **Early Career & Role Churn:** Employees in their first 1–3 years at the company or under a new manager exhibit higher turnover propensity.
- **Role Vulnerability:** Entry-level sales and lab roles require primary focus for employee retention strategies.